UE24CS341A – Software Engineering

Lab 7 – Fuzzing for Input Validation Bugs

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Section: E

SS1: python bike_share_rental_parser.py run on the buggy module

```
PS D:\Nihar\College stuff\Sem-5\SE\L6_L7_53_bike_share_rental\LAB7> py bike_share_rental_parser.py
==== Running parse_ride_duration ====
Enter a ride duration as HH:MM (e.g. 01:15): 01:15
Duration in minutes: 75

==== Running format_station_code ====
Enter a station code (e.g. a12): a12
Formatted station code: A-12

==== Running parse_ride_log ====
Enter a ride log line as type,minutes,subscriber (e.g. electric,45,yes): electric,45,yes
Parsed ride log: ('electric', 45, True)
```

SS2: pytest test_bike_share_rental_parser.py with completed tests failing on the buggy module

```
PS D:\Nihar\College stuff\Sem-5\SE\L6_L7_53_bike_share_rental\LAB7> py -m pytest test_bike_share_rental_parser.py -v
============================================= test session starts =============================================
platform win32 -- Python 3.11.0, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\Dell\AppData\Local\Programs\Python\Python311\python
.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: D:\Nihar\College stuff\Sem-5\SE\L6_L7_53_bike_share_rental\LAB7
plugins: anyio-4.10.0, hypothesis-6.170.15
collected 3 items

test_bike_share_rental_parser.py::test_parse_ride_duration_no_crash FAILED                    [ 33%]
test_bike_share_rental_parser.py::test_format_station_code_no_crash FAILED                    [ 66%]
test_bike_share_rental_parser.py::test_parse_ride_log_no_crash FAILED                         [100%]
```

SS3: python bike_share_rental_parser.py run on your fixed module

```
PS D:\Nihar\College stuff\Sem-5\SE\L6_L7_53_bike_share_rental\LAB7> py bike_share_rental_parser.py
Enter a ride duration as HH:MM (e.g. 01:15): 01:15
Duration in minutes: 75

==== Running format_station_code ====
Enter a station code (e.g. a12): a12
Formatted station code: A-12

==== Running parse_ride_log ====
Enter a ride log line as type,minutes,subscriber (e.g. electric,45,yes): electric,45,yes
Parsed ride log: ('electric', 45, True)
```

SS4: pytest test_bike_share_rental_parser.py with all tests passing on your fixed module

```
PS D:\Nihar\College stuff\Sem-5\SE\L6_L7_53_bike_share_rental\LAB7> py -m pytest test_bike_share_rental_parser.py -v
============================================= test session starts =============================================
platform win32 -- Python 3.11.0, pytest-8.3.4, pluggy-1.5.0 -- C:\Users\Dell\AppData\Local\Programs\Python\Python311\python
.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: D:\Nihar\College stuff\Sem-5\SE\L6_L7_53_bike_share_rental\LAB7
plugins: anyio-4.10.0, hypothesis-6.170.15
collected 3 items

test_bike_share_rental_parser.py::test_parse_ride_duration_no_crash PASSED                                [ 33%]
test_bike_share_rental_parser.py::test_format_station_code_no_crash PASSED                                [ 66%]
test_bike_share_rental_parser.py::test_parse_ride_log_no_crash PASSED                                     [100%]

============================================== 3 passed in 2.71s ==============================================
```